

Simple Harmonic Motion, solved

Step-by-step solutions to every practice problem in Chapter 9. Try each problem on your own before you read its solution.

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P9.1

No. SHM requires a restoring force proportional to the displacement from equilibrium, $F = -kx$. A bouncing ball feels a **constant** force, mg downward, during its flight, plus a brief, very large force during each impact. Its height follows a series of parabolic arcs, not a sinusoid.

There is another giveaway: its period depends on its amplitude. For a ball bouncing to height h , $T = 2\sqrt{2h/g}$. In SHM, the period is independent of the amplitude. The motion is *periodic*, but not *simple harmonic*.

P9.2

- Mass-spring: $T = 2\pi\sqrt{m/k}$ does not involve g , so the period stays 1.0 s.
- Pendulum: $T = 2\pi\sqrt{L/g}$, so $T \propto g^{-1/2}$:

$$T_{\text{Moon}} = 1.0 \times \sqrt{6} = 2.4 \text{ s.}$$

P9.3

- Speed is greatest at equilibrium ($x = 0$), where all the energy is kinetic: $v_{\text{max}} = A\omega$.
- Acceleration is greatest at the turning points ($x = \pm A$): $|a|_{\text{max}} = A\omega^2$.
- Same direction? Yes. The acceleration always points towards equilibrium. Whenever the particle is moving *towards* equilibrium, it is speeding up, and its velocity and acceleration point

the same way. When it is moving away from equilibrium, they point in opposite directions and it slows down.

P9.4

A marching column pushes on the bridge periodically, at the step frequency, roughly 2 Hz. If that frequency is close to one of the bridge's natural frequencies, each step adds energy in phase with the bridge's motion. The oscillation then grows towards the large resonant amplitude QF_0/k , which can overstress the structure. In 1831 the Broughton Suspension Bridge near Manchester collapsed under troops marching in step. Breaking step makes the individual footfalls incoherent, so their effects largely cancel instead of adding up.

P9.5

(a) Compare with $x = A \cos(\omega t + \phi)$:

- $A = 0.040$ m
- $\omega = 5\pi$ rad/s, so $f = \omega/2\pi = 2.5$ Hz and $T = 0.40$ s
- $\phi = \pi/3$ rad (60°)

(b) At $t = 0.10$ s, the phase is $5\pi(0.10) + \pi/3 = \pi/2 + \pi/3 = 5\pi/6$.

$$x = 0.040 \cos \frac{5\pi}{6} = 0.040(-0.866) = -0.0346 \text{ m.}$$

$$v = -A\omega \sin \frac{5\pi}{6} = -(0.040)(15.71)(0.5) = -0.314 \text{ m/s.}$$

$$a = -\omega^2 x = -(246.7)(-0.0346) = +8.5 \text{ m/s}^2.$$

At this moment the particle is on the negative side, moving further in the negative direction, while its acceleration points back towards equilibrium. It is slowing down.

P9.6

The angular frequency is $\omega = \sqrt{50/0.50} = 10$ rad/s.

(a) $E = \frac{1}{2}kA^2 = \frac{1}{2}(50)(0.080)^2 = 0.16$ J.

(b) $v_{\max} = A\omega = 0.080(10) = 0.80$ m/s.

$$(c) v = \omega \sqrt{A^2 - x^2} = 10 \sqrt{0.0064 - 0.0016} = 10(0.0693) = 0.69 \text{ m/s.}$$

Check by energy: $\frac{1}{2}mv^2 = 0.16 - \frac{1}{2}(50)(0.040)^2 = 0.16 - 0.04 = 0.12 \text{ J}$, so
 $v = \sqrt{2(0.12)/0.50} = 0.69 \text{ m/s. } \checkmark$

P9.7

(a) Parallel: $k = 100 + 300 = 400 \text{ N/m}$, so

$$T = 2\pi \sqrt{\frac{1.0}{400}} = 0.31 \text{ s.}$$

(b) Series: $\frac{1}{k} = \frac{1}{100} + \frac{1}{300} = \frac{4}{300}$, so $k = 75 \text{ N/m}$ and

$$T = 2\pi \sqrt{\frac{1.0}{75}} = 0.73 \text{ s.}$$

The series combination is softer than either spring alone.

P9.8

Spring constant: $k = mg/\Delta = 0.20(9.8)/0.040 = 49 \text{ N/m}$.

Period:

$$T = 2\pi \sqrt{\frac{m}{k}} = 2\pi \sqrt{\frac{0.20}{49}} = 0.40 \text{ s.}$$

Equivalently, $T = 2\pi \sqrt{\Delta/g} = 2\pi \sqrt{0.040/9.8} = 0.40 \text{ s. } \checkmark$

Maximum speed. The amplitude is 3.0 cm, measured from the new equilibrium position, and
 $\omega = 2\pi/T = 15.7 \text{ rad/s}$:

$$v_{\max} = A\omega = 0.030(15.7) = 0.47 \text{ m/s.}$$

P9.9

$$T = 2\pi \sqrt{\frac{L}{g}} = 2\pi \sqrt{\frac{1.0}{1.62}} = 4.9 \text{ s.}$$

On Earth it would be 2.0 s.

Measuring g : time many swings to find T accurately, measure L , then use

$$g = \frac{4\pi^2 L}{T^2}.$$

P9.10

Moment of inertia about the rim, by the parallel-axis theorem:

$$I = \frac{1}{2}MR^2 + MR^2 = \frac{3}{2}MR^2.$$

The distance from the pivot to the CM is $d = R$.

Period:

$$T = 2\pi\sqrt{\frac{I}{Mgd}} = 2\pi\sqrt{\frac{3R}{2g}} = 2\pi\sqrt{\frac{3(0.20)}{2(9.8)}} = 2\pi(0.175) = 1.1 \text{ s}.$$

Equivalent simple pendulum: $T = 2\pi\sqrt{L_{\text{eq}}/g}$ with $L_{\text{eq}} = I/(Md) = \frac{3}{2}R = 0.30 \text{ m}$.

P9.11

(a) **Damping rate.** Ten oscillations take $10(0.50) = 5.0 \text{ s}$. The amplitude decays as $e^{-\gamma t}$, so

$$e^{-5.0\gamma} = \frac{1}{2} \Rightarrow \gamma = \frac{\ln 2}{5.0} = 0.139 \text{ s}^{-1}.$$

(b) **Quality factor.** With such light damping, $\omega_0 \approx \omega_d = 2\pi/0.50 = 12.6 \text{ rad/s}$:

$$Q = \frac{\omega_0}{2\gamma} = \frac{12.6}{0.277} = 45.$$

(c) **Energy.** Energy is proportional to amplitude squared, so it falls to $(\frac{1}{2})^2 = \frac{1}{4}$, that is, 25% remains.

P9.12

(a) **Natural frequency:**

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{35\,000}{350}} = \frac{10}{2\pi} = 1.6 \text{ Hz}.$$

At speed v , the bumps arrive at a frequency v/d . Resonance occurs when $v/d = f_0$:

$$v = f_0 d = 1.59(10) = 15.9 \text{ m/s} \approx 57 \text{ km/h}.$$

(b) At that speed, each bump arrives just as the car body completes one bounce, so energy is pumped in on every cycle and the bouncing builds up. Slower or faster, the bumps arrive out of step with the bounce and partly cancel its effect.

(c) Damping limits the resonant amplitude to about Q times the static response. With Q near 1–2 (around 30% of critical damping), the resonance is broad and mild, so no speed is very uncomfortable. Worn-out shock absorbers raise Q , and the car then wallows badly at the resonant speed.

P9.13

(a) **Equilibrium:**

$$U'(x) = U_0 \left(-\frac{a}{x^2} + \frac{1}{a} \right) = 0 \Rightarrow x^2 = a^2 \Rightarrow x_0 = a \quad (x > 0).$$

Stability:

$$U''(x) = \frac{2U_0 a}{x^3} \Rightarrow U''(a) = \frac{2U_0}{a^2} > 0.$$

It is a minimum, so the equilibrium is stable.

(b) **Small oscillations.** Near the minimum, $U \approx U(a) + \frac{1}{2}U''(a)(x - a)^2$, which is a spring with $k_{\text{eff}} = 2U_0/a^2$. So

$$\omega = \sqrt{\frac{k_{\text{eff}}}{m}} = \sqrt{\frac{2U_0}{ma^2}}.$$

P9.14

Energy method. When one side is up by x and the other down by x , it is as if a column of liquid of length x (mass $\rho A x$) has been moved from the low side to the high side, raising its centre of mass by x . So

$$U = (\rho A x) g x = \rho A g x^2.$$

The whole column, of mass ρAL , moves with speed $\dot{x}$:

$$K = \frac{1}{2} \rho AL \dot{x}^2.$$

Energy conservation:

$$\frac{1}{2} \rho AL \dot{x}^2 + \rho Ag x^2 = \text{const.}$$

Differentiate with respect to time and divide by $\rho A \dot{x}$:

$$L\ddot{x} + 2gx = 0 \Rightarrow \ddot{x} = -\frac{2g}{L}x.$$

This is SHM with $\omega = \sqrt{2g/L}$. ✓

Force method (check). The unbalanced column of height $2x$ has weight $2\rho Agx$, which acts to restore the level. This force accelerates the whole mass ρAL : $\rho AL \ddot{x} = -2\rho Agx$. Same result.

Evaluate. The frequency does not depend on the density or the tube's cross-section, only on the column length. This is the basis of U-tube manometer dynamics, and of the slow “sloshing” modes of liquid in tanks.

P9.15

The fractional change in period between the two amplitudes is

$$\frac{\Delta T}{T} \approx \frac{\theta_2^2 - \theta_1^2}{16} = \frac{(0.1745)^2 - (0.0873)^2}{16} = \frac{0.03046 - 0.00762}{16} = 1.43 \times 10^{-3}.$$

The period is **longer**, so the clock runs **slow**:

$$\text{loss} = 1.43 \times 10^{-3} \times 86\,400 \text{ s} = 123 \text{ s per day},$$

about 2 minutes a day. Precision clocks keep their amplitude small and constant for exactly this reason.

P9.16

(a) **Derivation.** Substitute $x = A \cos(\omega t - \delta)$ into $m\ddot{x} + b\dot{x} + kx = F_0 \cos \omega t$:

$$A(k - m\omega^2) \cos(\omega t - \delta) - Ab\omega \sin(\omega t - \delta) = F_0 \cos \omega t.$$

Write the right-hand side in terms of $(\omega t - \delta)$:

$$F_0 \cos \omega t = F_0 [\cos(\omega t - \delta) \cos \delta - \sin(\omega t - \delta) \sin \delta].$$

Matching the coefficients of $\cos(\omega t - \delta)$ and $\sin(\omega t - \delta)$:

$$A(k - m\omega^2) = F_0 \cos \delta, \quad Ab\omega = F_0 \sin \delta.$$

Squaring and adding the two equations:

$$A = \frac{F_0}{\sqrt{(k - m\omega^2)^2 + b^2\omega^2}} = \frac{F_0/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + (2\gamma\omega)^2}}. \checkmark$$

Dividing them:

$$\tan \delta = \frac{b\omega}{k - m\omega^2} = \frac{2\gamma\omega}{\omega_0^2 - \omega^2}.$$

(b) **Peak.** A is largest where $D = (\omega_0^2 - \omega^2)^2 + 4\gamma^2\omega^2$ is smallest. Treat D as a function of $u = \omega^2$:

$$\frac{dD}{du} = -2(\omega_0^2 - u) + 4\gamma^2 = 0 \Rightarrow u = \omega_0^2 - 2\gamma^2 \Rightarrow \omega_r = \sqrt{\omega_0^2 - 2\gamma^2}.$$

For light damping, $\omega_r \approx \omega_0$.

(c) At $\omega = \omega_0$: the denominator of $\tan \delta$ vanishes, so $\delta = 90^\circ$. The displacement lags the force by a quarter cycle. The velocity is

$$\dot{x} = -A\omega \sin(\omega t - 90^\circ) = A\omega \cos \omega t,$$

which is exactly **in phase** with the force. The instantaneous power, $P = F\dot{x} = F_0 A \omega \cos^2 \omega t$, is then never negative: the driver always pushes in the direction of motion. That is the condition for the greatest energy transfer, which is why the power absorbed peaks exactly at ω_0 . This is what you do instinctively when pushing a child on a swing.

P9.17

(a) **Collision** (perfectly inelastic, so momentum is conserved):

$$V = \frac{0.25(5.0)}{1.25} = 1.0 \text{ m/s}.$$

(b) **Oscillation.** The new mass is 1.25 kg:

$$\omega = \sqrt{\frac{100}{1.25}} = 8.94 \text{ rad/s}, \quad T = \frac{2\pi}{\omega} = 0.70 \text{ s}.$$

The block starts at equilibrium with speed V , so $V = v_{\max} = A\omega$:

$$A = \frac{V}{\omega} = \frac{1.0}{8.94} = 0.11 \text{ m}.$$

(c) **Energy lost.**

$$K_{\text{before}} = \frac{1}{2}(0.25)(5.0)^2 = 3.125 \text{ J}, \quad K_{\text{after}} = \frac{1}{2}(1.25)(1.0)^2 = 0.625 \text{ J}.$$

80% is lost, which equals $M/(m + M) = 1.0/1.25$, as for any perfectly inelastic collision with a body at rest.

P9.18

(a) **Scaling.** The swing period is $T \propto \sqrt{L/g}$, so the cadence (steps per second) is proportional to $1/T \propto \sqrt{g/L}$. With step length $\propto L$:

$$v \propto L\sqrt{\frac{g}{L}} = \sqrt{gL}.$$

(b) **Child versus adult.** With L halved:

$$\frac{v_{\text{child}}}{v_{\text{adult}}} = \sqrt{\frac{1}{2}} = 0.71.$$

The child walks naturally at about 70% of the adult's speed, with a faster cadence. Children trot to keep up.

(c) **On the Moon, with $g/6$:**

$$v_{\text{Moon}} = \frac{v_{\text{Earth}}}{\sqrt{6}} \approx 0.4 v_{\text{Earth}} \approx 0.5 \text{ m/s}.$$

Natural walking becomes painfully slow. Biomechanists use the dimensionless *Froude number*, $Fr = v^2/gL$. Humans and animals switch from walking to running at about $Fr \approx 0.5$. On the Moon that happens at only $v = \sqrt{0.5(1.62)(0.9)} \approx 0.85 \text{ m/s}$. So to move at any useful speed, astronauts had to adopt a bounding, hopping gait, exactly as the Apollo films show.

All worked examples and practice problems in these notes are original.
Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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